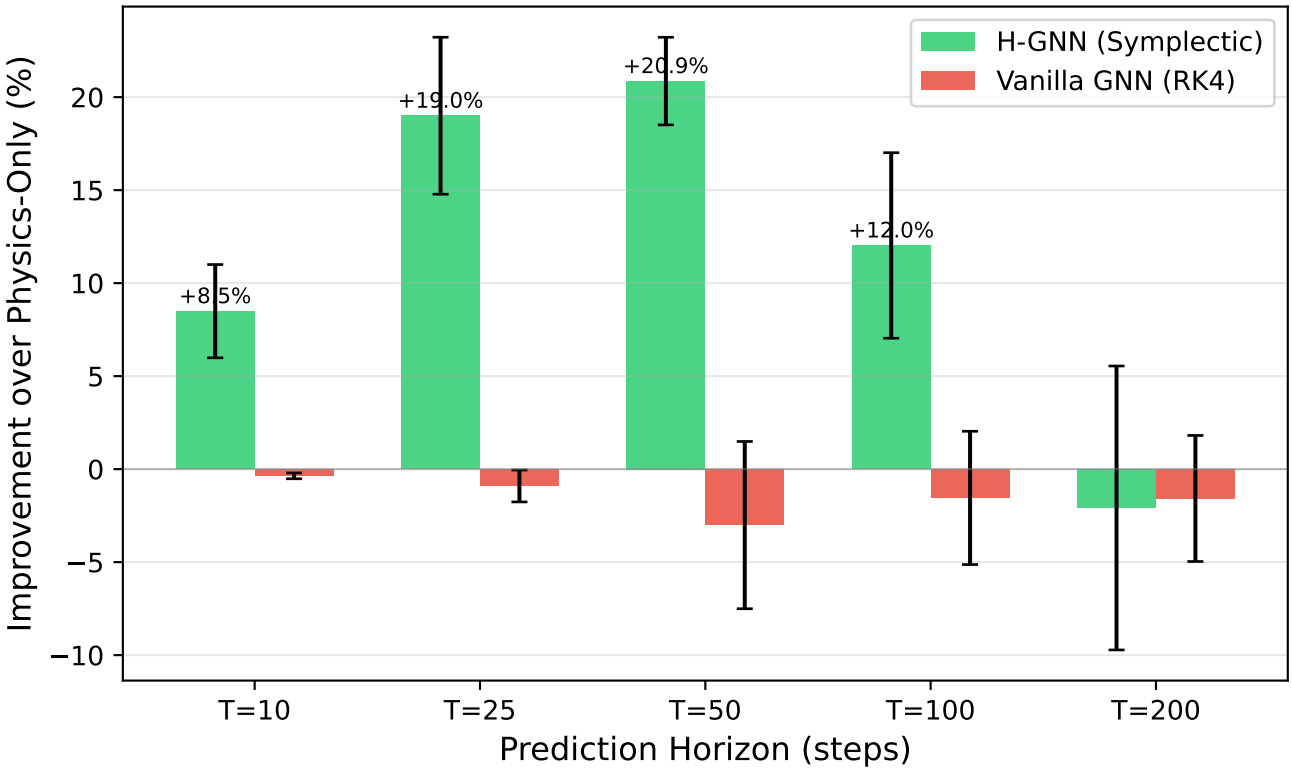
Symplectic Structure Enables Learning
H-GNN (Symplectic)
Vanilla GNN (RK4)
Improvement over Physics-Only (%)
+8.5%
+19.0%
+20.9%
+12.0%
20
15
10
5
0
−5
−10
T=10
T=25
T=50
T=100
T=200
Prediction Horizon (steps)